Creating Microservices for account and loan

In this hands on exercises, we will create two microservices for a bank. One microservice for handing accounts and one for handling loans.

Each microservice will be a specific independent Spring RESTful Webservice maven project having it's own pom.xml. The only difference is that, instead of having both account and loan as a single application, it is split into two different applications. These webservices will be a simple service without any backend connectivity.

Follow steps below to implement the two microservices:

Account Microservice

- Create folder with employee id in D: drive
- Create folder named 'microservices' in the new folder created in previous step. This folder will contain all the sample projects that we will create for learning microservices.
- Open https://start.spring.io/ in browser
- Enter form field values as specified below:
  - **Group:** com.cognizant
  - **Artifact:** account
- Select the following modules
  - Developer Tools > Spring Boot DevTools
  - Web > Spring Web
- Click generate and download the zip file
- Extract 'account' folder from the zip and place this folder in the 'microservices' folder created earlier
- Open command prompt in account folder and build using mvn clean package command
- Import this project in Eclipse and implement a controller method for getting account details based on account number. Refer specification below:
  - Method: GET
  - Endpoint: /accounts/{number}
  - Sample Response. Just a dummy response without any backend connectivity.

```
{ number: "00987987973432", type: "savings", balance: 234343 }
```

- Launch by running the application class and test the service in browser

Loan Microservice

- Follow similar steps specified for Account Microservice and implement a service API to get loan account details
  - Method: GET
  - Endpoint: /loans/{number}
  - Sample Response. Just a dummy response without any backend connectivity.

```
{ number: "H00987987972342", type: "car", loan: 400000, emi: 3258, tenure:
18 }
```

- Launching this application by having account service already running
- This launch will fail with error that the bind address is already in use
- The reason is that each one of the service is launched with default port number as 8080. Account service is already using this port and it is not available for loan service.
- Include "server.port" property with value 8081 and try launching the application
- Test the service with 8081 port

Now we have two microservices running on different ports.

NOTE: The console window of Eclipse will have both the service console running. To switch between different consoles use the monitor icon within the console view.

Week 4 – Creating Microservices for Account and Loan

Objective

To create two independent Spring Boot REST microservices:

- Account Microservice
- Loan Microservice

Each service runs independently on a different port and returns dummy JSON data without backend connectivity.

Technologies Used

- Java 17
- Spring Boot 4
- Spring Web

- Spring Boot DevTools
- Maven
- IntelliJ IDEA

Project Structure

```
microservices
│
├── account
│
└── loan
```

Account Microservice

Endpoint

GET /accounts/{number}

URL

http://localhost:8080/accounts/00987987973432

Sample Response

```
{
  "number": "00987987973432",
  "type": "savings",
  "balance": 234343.0
}
```

Loan Microservice

Endpoint

GET /loans/{number}

URL

http://localhost:8081/loans/H00987987972342

Sample Response

{

"number": "H00987987972342",

"type": "car",

"loan": 400000.0,

"emi": 3258,

"tenure": 18

}

Port Configuration

By default, Spring Boot applications run on **8080**.

The Loan microservice was configured to use:

server.port=8081

to avoid port conflicts.

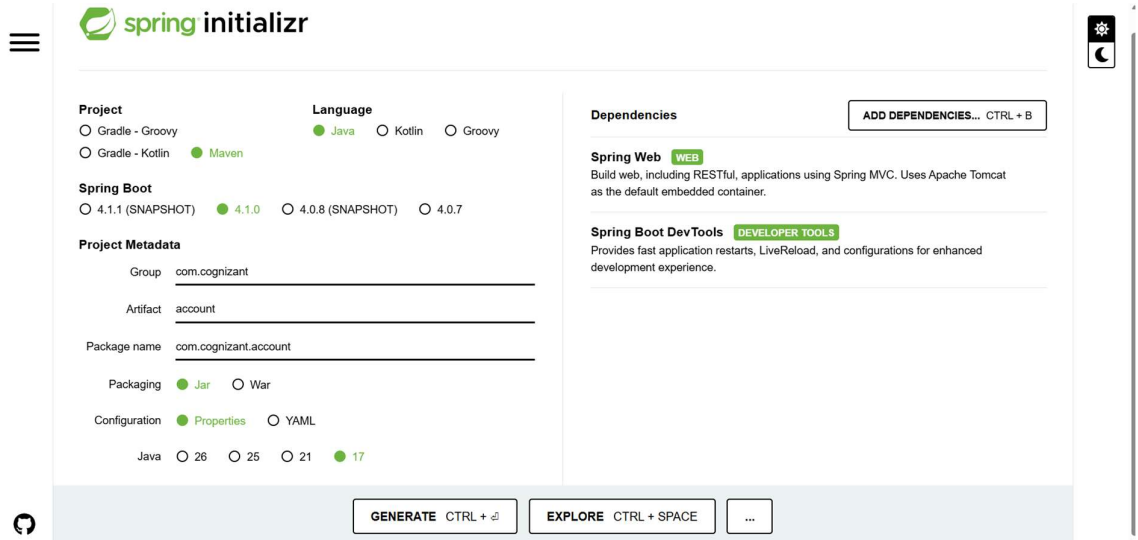
Result

Successfully created and tested two independent Spring Boot microservices.

- Account Service running on Port **8080**
- Loan Service running on Port **8081**

Both REST APIs returned the expected JSON responses.

p


spring initializr
Project
Gradle - Groovy
Gradle - Kotlin
Maven
Language
Java
Kotlin
Groovy
Spring Boot
4.1.1 (SNAPSHOT)
4.1.0
4.0.8 (SNAPSHOT)
4.0.7
Project Metadata
Group
com.cognizant
Artifact
account
Package name
com.cognizant.account
Packaging
Jar
War
Configuration
Properties
YAML
Java
26
25
21
17
Dependencies
ADD DEPENDENCIES... CTRL + B
Spring Web
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GENERATE CTRL + ⏎
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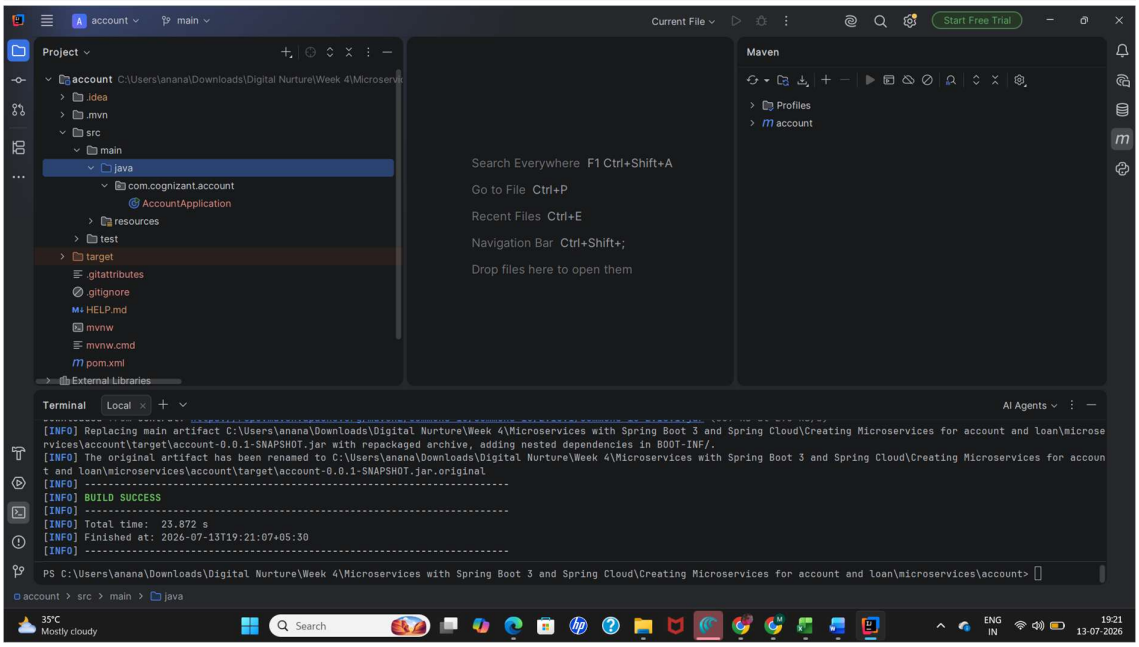
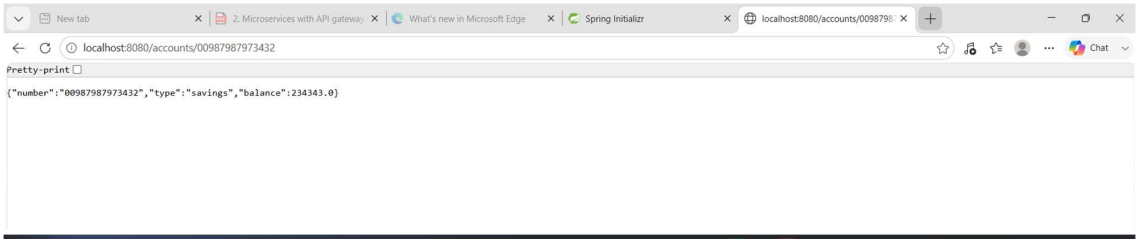
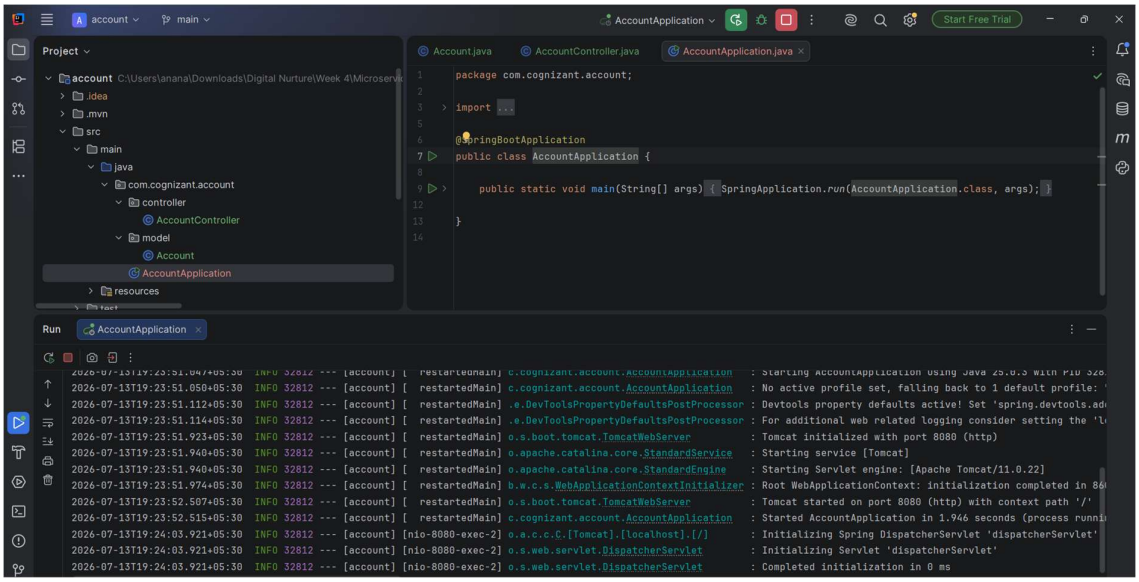
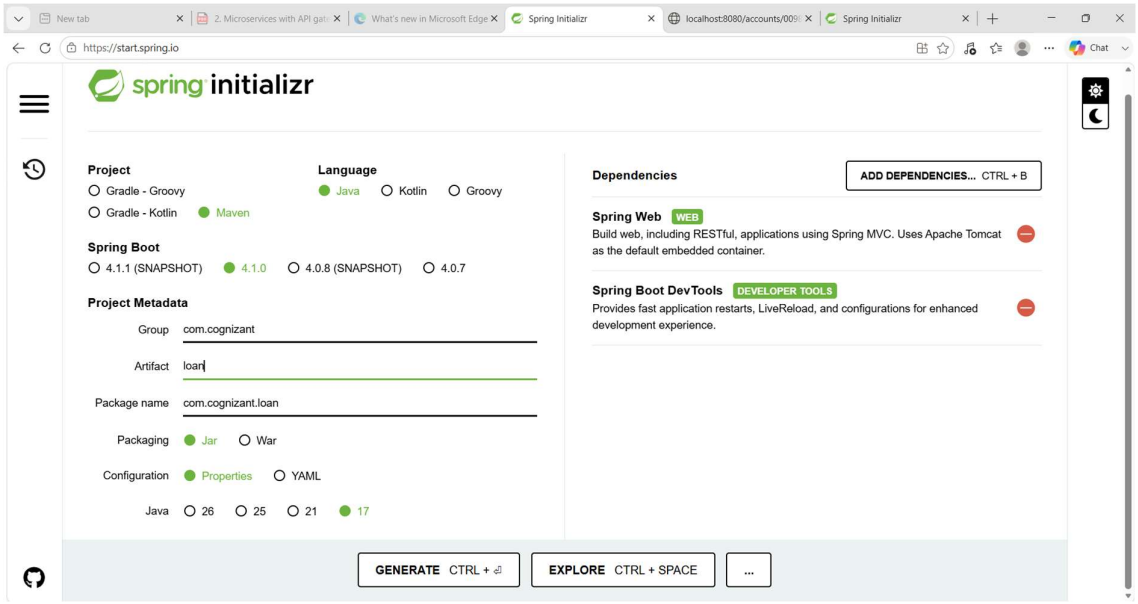

account
main
Current File
Start Free Trial
Project
account C:\Users\anana\Downloads\Digital Nurture\Week 4\Microservic
.idea
.mvn
src
main
java
com.cognizant.account
AccountApplication
resources
test
target
.gitattributes
.gitignore
HELP.md
mvnw
mvnw.cmd
pom.xml
External Libraries
Search Everywhere F1 Ctrl+Shift+A
Go to File Ctrl+P
Recent Files Ctrl+E
Navigation Bar Ctrl+Shift+;
Drop files here to open them
Maven
Profiles
account
Terminal
Local
AI Agents
[INFO] Replacing main artifact C:\Users\anana\Downloads\Digital Nurture\Week 4\Microservices with Spring Boot 3 and Spring Cloud\Creating Microservices for account and loan\microservices\account\target\account-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\anana\Downloads\Digital Nurture\Week 4\Microservices with Spring Boot 3 and Spring Cloud\Creating Microservices for account and loan\microservices\account\target\account-0.0.1-SNAPSHOT.jar.original
[INFO] --
[INFO] BUILD SUCCESS
[INFO] --
[INFO] Total time: 23.872 s
[INFO] Finished at: 2026-07-13T19:21:07+05:30
[INFO] --
PS C:\Users\anana\Downloads\Digital Nurture\Week 4\Microservices with Spring Boot 3 and Spring Cloud\Creating Microservices for account and loan\microservices\account>
account > src > main > java
35°C
Mostly cloudy
Search
ENG
IN
19:21
13-07-2026


New tab
2. Microservices with API gateway
What's new in Microsoft Edge
Spring Initializr
localhost:8080/accounts/0098798
localhost:8080/accounts/00987987973432
Chat
Pretty-print
{"number":"00987987973432","type":"savings","balance":234343.0}


account
main
AccountApplication
Start Free Trial
Project
account C:\Users\anana\Downloads\Digital Nurture\Week 4\Microservic
.idea
.mvn
src
main
java
com.cognizant.account
controller
AccountController
model
Account
AccountApplication
resources
Account.java
AccountController.java
AccountApplication.java
package com.cognizant.account;
import ...
@SpringBootApplication
public class AccountApplication {
public static void main(String[] args) { SpringApplication.run(AccountApplication.class, args); }
}
Run
AccountApplication
2026-07-13T19:23:51.050+05:30 INFO 32812 --- [account] [restartedMain] c.cognizant.account.AccountApplication : No active profile set, falling back to 1 default profile:
2026-07-13T19:23:51.112+05:30 INFO 32812 --- [account] [restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.ad
2026-07-13T19:23:51.114+05:30 INFO 32812 --- [account] [restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'l
2026-07-13T19:23:51.923+05:30 INFO 32812 --- [account] [restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-13T19:23:51.940+05:30 INFO 32812 --- [account] [restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-13T19:23:51.940+05:30 INFO 32812 --- [account] [restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-13T19:23:51.974+05:30 INFO 32812 --- [account] [restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 86
2026-07-13T19:23:52.507+05:30 INFO 32812 --- [account] [restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-13T19:23:52.515+05:30 INFO 32812 --- [account] [restartedMain] c.cognizant.account.AccountApplication : Started AccountApplication in 1.946 seconds (process runni
2026-07-13T19:24:03.921+05:30 INFO 32812 --- [account] [nio-8080-exec-2] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-13T19:24:03.921+05:30 INFO 32812 --- [account] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-13T19:24:03.921+05:30 INFO 32812 --- [account] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms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Spring Boot
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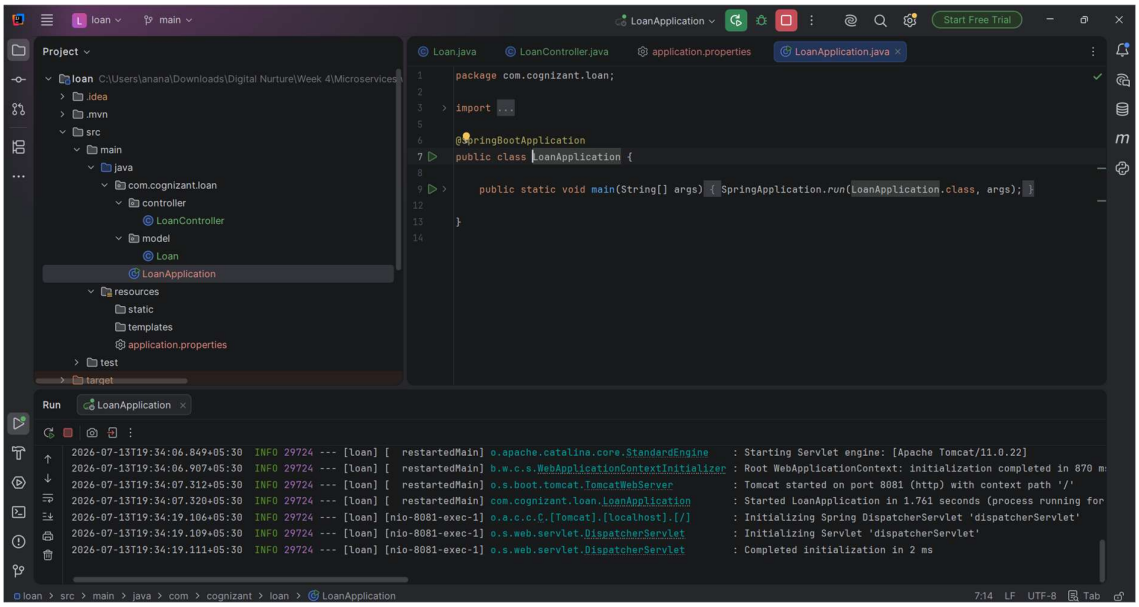
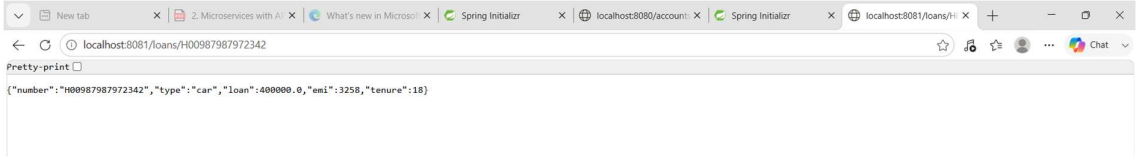

Project
loan
.idea
.mvn
src
main
java
com.cognizant.loan
controller
LoanController
model
Loan
LoanApplication
resources
static
templates
application.properties
test
target
Loan.java
LoanController.java
application.properties
LoanApplication.java
package com.cognizant.loan;
import ...
@SpringBootApplication
public class LoanApplication {
public static void main(String[] args) { SpringApplication.run(LoanApplication.class, args); }
}
Run
LoanApplication
2026-07-13T19:34:06.849+05:30 INFO 29724 --- [loan] [restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-13T19:34:06.907+05:30 INFO 29724 --- [loan] [restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 870 ms
2026-07-13T19:34:07.312+05:30 INFO 29724 --- [loan] [restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path '/'
2026-07-13T19:34:07.320+05:30 INFO 29724 --- [loan] [restartedMain] com.cognizant.loan.LoanApplication : Started LoanApplication in 1.761 seconds (process running for
2026-07-13T19:34:19.106+05:30 INFO 29724 --- [loan] [nio-8081-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-13T19:34:19.109+05:30 INFO 29724 --- [loan] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-13T19:34:19.111+05:30 INFO 29724 --- [loan] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms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Pretty-print
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